How do diffusion-trained tokens compare chemically?

Diffusion scores lower than both JEPA checkpoints on the shared final-token chemical readout.

0.7244

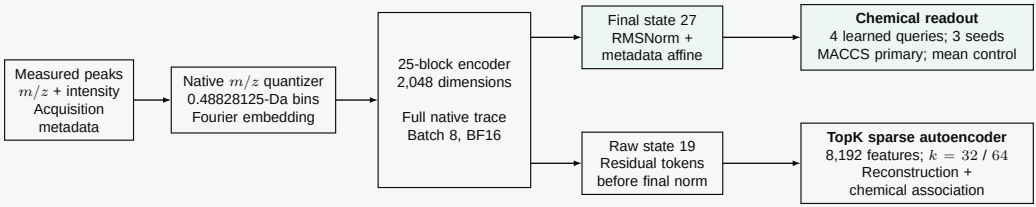
Final-token attention MACCS AUROC; diffusion checkpoint 831,000

A matched token-level evaluation; training exposure and native input quantization still differ.

Latest committed endpoint of adiff-1p8b-ep-v6e32-4d-20260922-080322, pinned by 60 object generations. Comparison arms: frozen-teacher JEPA 1,190k and periodic-teacher JEPA 1,209k.

Preserve peak tokens for chemical readouts and sparse analysis

a Native diffusion encoder, frozen during downstream fitting



Clean-input encoder only; diffusion predictor and categorical reconstruction head are outside the chemical evaluation.

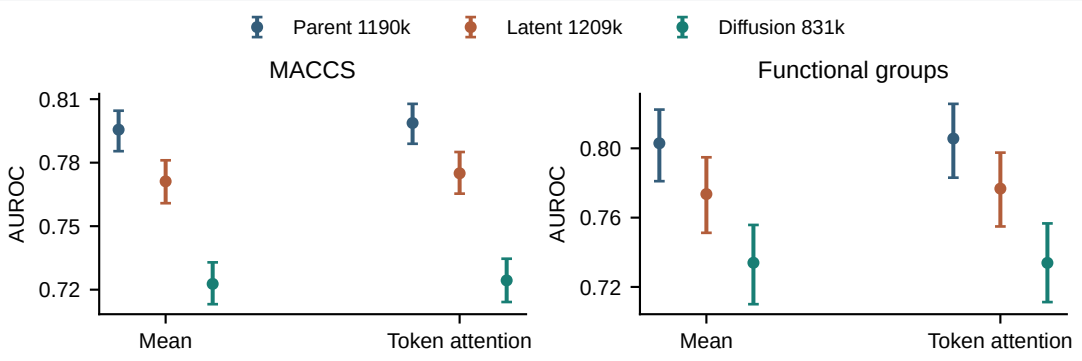
Valid physical peak tokens only; CLS and padding excluded. Readouts use labels; SAE reconstruction training does not.

Arrows: forward computation or activation export. Train / validation / test: 4,096 / 1,024 / 1,024 independent molecular connectivities.

Token attention is the primary chemistry endpoint; sparse reconstruction measures a different property.

One representative spectrum per molecular connectivity: 4,096 train / 1,024 validation / 1,024 test. Native JAX BF16, batch 8; physical peaks exclude CLS and padding. Exact native/instrumented/cache parity required.

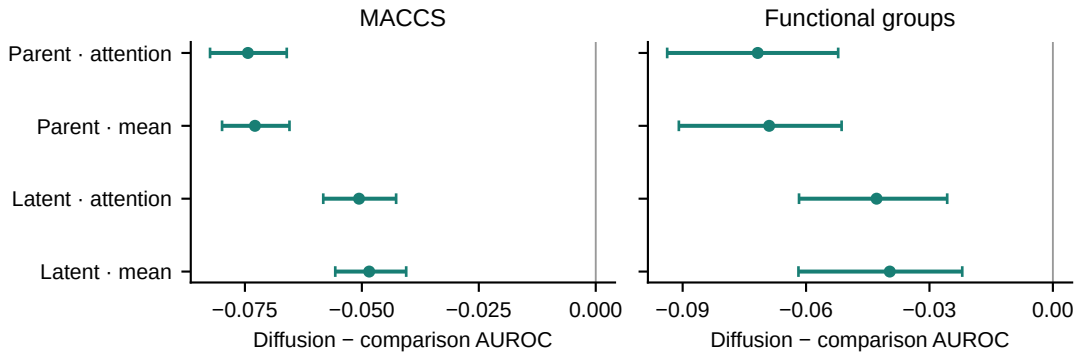
Diffusion trails the final-token chemical readout



Diffusion attention gains +0.0016 MACCS AUROC over mean; its paired interval includes zero.

Final state 27; matched physical-token mean control and learned attention. AUROC of mean logits across three independently fitted seeds. Shared fixed support: 145 MACCS / 13 functional-group targets; 1,024 test molecules.

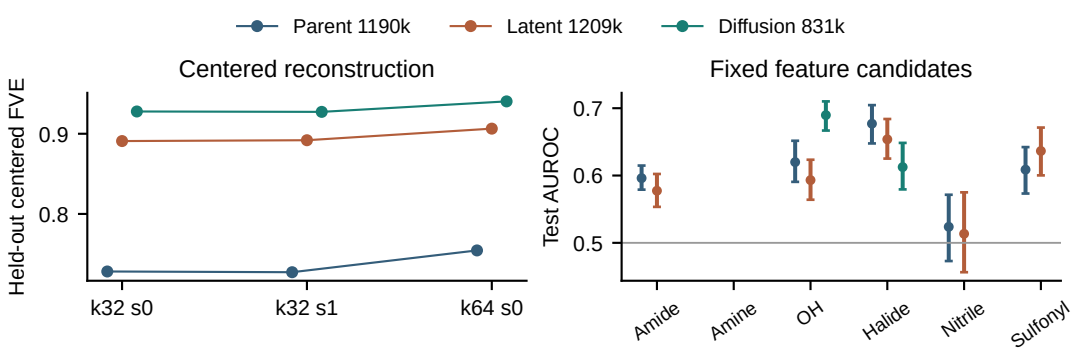
Paired molecules separate the checkpoint differences



Attention differences: parent -0.0744; periodic -0.0506 MACCS AUROC.

Diffusion minus comparison; positive favors diffusion. Shared molecule-bootstrap counts; 980 of 1,000 MACCS resamples retain both classes for every fixed target. Undefined replicates omitted. Descriptive 95% intervals; no pretraining-seed uncertainty or multiplicity correction.

Sparse reconstruction and chemical recognition are separate



Diffusion k=32 seed0: 92.8% held-out variance explained; 3 chemical candidates frozen before test.

Raw19, 8,192 features; train-only normalization and equal molecule weights. FVE is relative to the training mean; original-unit relative L2 is 0.259 for diffusion versus 0.088 / 0.054 for parent / periodic. Right: all frozen candidates per concept, paired molecule 95% intervals; missing candidates remain gaps. No diffusion feature interventions.

Shared dataset, distinct training histories and native inputs

Training exposure

All three logs begin at step 0.
Nominal spectrum presentations:
Parent: 1.219 billion
Periodic: 1.238 billion
Diffusion: 0.851 billion

Native input difference

Diffusion quantizes m/z into 0.48828125-Da bin midpoints before Fourier embedding.

JEPA arms retain continuous m/z. The loss alone is not isolated.

Compare the measured endpoints; do not attribute the difference solely to the diffusion objective.

Pinned checkpoint contracts share dataset revision, manifest and shard-plan hashes. Nominal presentations = global step × configured global batch 1,024, not unique spectra; accumulation is already included. Teacher histories are not added.

A bounded checkpoint comparison, with reusable artifacts

Checkpoint provenance
Run config, latest committed object generations, CRC32C and lineage evidence; no parent initialization is specified for diffusion.

Chemical features need nuisance controls
The strongest hydroxyl candidate (f4682) has test AUROC 0.744, but also mass correlation 0.599; strict matched-pair ordering is 0.517.

Claim boundaries
No matched training-budget experiment, pretraining-seed replication, diffusion feature ablation or validated peak-to-fragment assignments. Readout scores are not calibrated chemical probabilities.

Token-level readouts answer chemical accessibility; SAE structure adds a separate, qualified view.

Artifacts: artifacts/encoder_chemistry_20260930/diffusion_extension/. Analysis: scripts/analyze_encoder_chemistry_diffusion.py. Source and claim hashes: research/evidence_index.json. Full historical model source is not recoverable from W&B's train.py alone.